3Consulting
BUILD TRUST IN THE DIGITAL ECONOMY

ENTERPRISE DATA GOVERNANCE · CASE STUDY

SafeData: Automated NDPA & GDPR Consent Disposal

BI Scientist • AI/ML Engineer • Python Developer

Data-retention and consent-withdrawal platform: retention schedules, legal holds, dual approval for high-risk records, and a SHA-256 hash-chained audit log. Deployment pending verification.

THE COMPLIANCE CHALLENGE

Why Enterprise Data Retention Breaks

NDPA Section 34 and GDPR Article 17 require verifiable, irreversible destruction of customer data upon consent withdrawal.

Orphaned PII in Backups

User consent is withdrawn in CRM, but records persist in analytics replicas and backups.

No automated system maps customer PII across disparate microservices.

Exposure to NDPA and GDPR enforcement.

Manual Spreadsheet Failure

DPOs rely on spreadsheets to track departmental data retention schedules.

No dual-approval or legal-hold check before a deletion.

Audit trails are hard to verify after the fact.

Unprovable Deletion

Organisations need evidence of what was disposed of, when and why.

A plain DELETE leaves no tamper-evident record.

SafeData records each disposal in a hash-chained log.

SYSTEM ARCHITECTURE

Automated Scanners, Retention Engine & Hash-Chain Audit

Data-lifecycle workflow built on FastAPI and PostgreSQL with tenant scoping.

Intake & Webhooks

- Scoped API keys & HMAC signatures.
- Listens for consent-withdrawn events.
- Maps PII identifiers across client tables.

Retention Engine

- Jurisdiction-aware retention schedules.
- Consent-withdrawal disposal countdowns.
- Dual-approval workflows & legal holds.

Tamper-Proof Audit

- SHA-256 chained audit entries.
- Chain can be re-verified on demand.
- Disposal evidence for DPO review.

PRODUCT WALKTHROUGH

SafeData Governance Console

DPO workspace for disposal queues, RoPA registers and legal-hold approvals (demo data).

- **Disposal Pipeline:** Records move from active use through review and approval to disposal, with each step logged.
- **Dual-Key Approvals:** Legal & Compliance officers must co-sign high-sensitivity dataset deletions.
- **RoPA Register:** Record of Processing Activities with processor entries, alongside DSAR, DPIA and breach workflows.

FastAPI | PostgreSQL | Docker | NDPA / GDPR



safedata.3consult-ng.com/app/retention

WHERE IT STANDS

Built, Tested, Not Yet Launched

What exists in the codebase today. No customer results are claimed.

Status
Pre-launch
Deployment pending verification

Audit Log
SHA-256
Hash-chained, verifiable

Access
RBAC + 2FA
Admin, DPO, data owner roles

CI
Tests
Plus dependency and secret scans

Designed Around the Rules

- Workflows map to NDPA 2023 and GDPR Article 17 (right to erasure).
- Supports compliance work; it is not a compliance certificate.
- Tamper-evident logs give DPOs evidence to review.

Engineering

- Legal holds override ordinary disposal timing.
- Tenant-scoped PostgreSQL data model.
- Background worker runs scheduled scans and notifications.

CASE STUDY TAKEAWAY

Keep what you must. Release what you should.

SafeData turns retention and consent-withdrawal rules into a tracked, auditable workflow.

Company: 3Consulting Limited (Lagos, Nigeria)
Planned site: safedata.3consult-ng.com (deployment pending verification)